

NeuroShield: A Personalized, Multi-Signal System for Honest Real-Time Stress Detection

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Abstract

I present NeuroShield, a system that estimates a person’s physiological stress in real time from low-cost wearable-grade sensors, and does so *relative to that individual’s own resting baseline* rather than a population average. The motivating observation is that absolute physiological values are ambiguous across people: two individuals can both be fully at rest at 58 and 82 beats per minute, so a model fed raw values expends capacity learning *who* a person is instead of *how aroused* they are. NeuroShield calibrates to each user during a short quiet period and expresses every subsequent reading as a deviation from that baseline. Every 60 seconds it derives 36 features from four physiological channels—photoplethysmography (PPG), electrodermal activity (EDA), skin temperature, and accelerometry—and feeds them to a two-headed gradient-boosted model: a calibrated head that emits a 0–100 graded stress index, and a four-class head that separates stress arousal from positive arousal such as amusement. Under grouped leave-one-subject-out (LOSO) evaluation—training on some subjects and testing only on subjects never seen during training—the graded-stress head attains a balanced accuracy of **0.919**, exceeding the strongest comparable wrist-only result reported in the literature (0.874). Beyond raw accuracy, the system is designed for *epistemic honesty*: it abstains from prediction when motion or poor signal quality render a reading untrustworthy, it surfaces its own uncertainty (a ± 0.036 standard error and a per-subject accuracy range of 0.50–1.00), and it is structurally prevented from issuing any clinical or diagnostic claim. I describe the research, problem, objectives, plan, methodology, datasets, and results behind the system, and I report a negative result establishing that 0.919 is the practical ceiling for the available data.

Keywords: affective computing, stress detection, wearable sensors, heart-rate variability, electrodermal activity, personalization, gradient boosting, model calibration, leave-one-subject-out validation.

1 Introduction

Chronic stress is among the most prevalent and costly health burdens, yet the instruments used to track it are weak. The dominant tool—self-report—is unreliable: people are poor at noticing their own stress, and by the time they consciously register it they have often been under load for hours. The autonomic nervous system, however, responds first, and it responds *measurably*. Psychological stress activates the sympathetic branch of the autonomic nervous system, producing a consistent peripheral signature: elevated heart rate, suppressed heart-rate variability, increased sweat-gland activity, and cooling of the skin as blood is redirected toward the core. Each of these is observable with inexpensive sensors.

The difficulty is that turning those signals into a trustworthy estimate of stress is far harder than it first appears, for three reasons. First, physiological signals vary enormously between individuals, so a fixed rule (“heart rate above 90 $\Rightarrow$ stress”) is meaningless across a population. Second, wearable sensors on a moving body produce unreliable data much of the time, and a model that outputs a confident number during that period is not merely useless but actively

misleading. Third, the commercial temptation is to claim detection of “anxiety” or “burnout,” which the underlying signals physically cannot support.

NeuroShield addresses all three. My contributions are:

1. A **personalization method** that expresses every feature relative to each user’s own calibrated baseline, which I show is the single largest driver of cross-subject accuracy.
2. A **two-headed model** that produces a calibrated, graded (0–100) stress index and additionally distinguishes stress arousal from positive arousal.
3. An **abstention mechanism** that makes refusal to predict a first-class output.
4. An **honest evaluation**: LOSO throughout, a reported error bar, held-out datasets, and a documented negative result showing that further tuning does not improve on 0.919.
5. An **end-to-end streaming system** in which hardware acts only as a data source, so the entire pipeline is validated before any device is attached.

2 Background and Related Work

I grounded the design in a review of physiological stress sensing, from which three findings shaped every subsequent decision.

Stress has a measurable, multi-channel signature. The sympathetic response elevates heart rate and reduces beat-to-beat variability, because the parasympathetic (vagal) “brake” on the heart is released. It increases eccrine sweat-gland activity, raising skin conductance. And it drives peripheral vasoconstriction, cooling the skin at the extremities. A single channel is ambiguous; the combination is informative. Electrodermal activity is of particular value because eccrine sweat glands are innervated *solely* by the sympathetic branch, with no antagonistic parasympathetic input [3], making EDA the cleanest available proxy for arousal.

Reported accuracies are frequently inflated by weak validation. Many studies report 93–99% accuracy, but these commonly rely on k -fold or intra-subject cross-validation, which allows the same individual to appear in both training and test partitions. A model can then memorize person-specific idiosyncrasies rather than learn a generalizable mapping. The honest protocol is *leave-one-subject-out* (LOSO), in which the test subject is entirely unseen during training. Under LOSO, the strongest comparable wrist-only result I identified was 0.874 [2]. I adopted LOSO as the sole evaluation metric.

Personalization is the dominant lever, and naive data pooling backfires. The wearable affect literature, beginning with the WESAD benchmark [1], repeatedly identifies subject-relative normalization as the largest source of cross-subject gain. I further found—and later empirically confirmed—that merely concatenating datasets to obtain “more data” can *reduce* accuracy, because it folds cross-protocol distribution shift into the model without domain adaptation.

3 Problem Formulation and Objectives

3.1 Problem statement

Given a stream of physiological samples from a small set of low-cost sensors, produce an honest, interpretable, real-time estimate of a person’s stress that (i) adapts to individual physiology, (ii) recognizes and abstains under untrustworthy signal conditions, and (iii) never over-claims beyond what the sensors can support.

Table 1: Project timeline (Gantt chart). Shaded cells indicate active weeks.

Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1. Background research & scoping	■	■								
2. Data contracts & dataset loaders	■	■	■							
3. Feature pipeline (v1)		■	■	■						
4. Baseline model (M1) + LOSO			■	■	■					
5. Runtime: calibration, gate, states, explanations				■	■	■	■			
6. Backend + dashboard (v1)					■	■	■	■		
7. Dataset harmonization + held-out validation						■	■	■	■	
8. Accuracy improvement (features-v2, personalization)							■	■	■	■
9. Real streaming + interface rebuild								■	■	■
10. Documentation, spec, hardening									■	■

3.2 Objectives

I defined the project against seven concrete objectives:

1. Build a complete software-first pipeline (data $\rightarrow$ features $\rightarrow$ model $\rightarrow$ runtime $\rightarrow$ interface) that runs from replayed or streamed data with no hardware dependency.
2. Detect stress in a personalized manner, calibrated to each user’s own baseline.
3. Match or exceed the ~ 0.874 LOSO benchmark under honest evaluation.
4. Produce a graded, interpretable output—a 0–100 index with a plain-language explanation of which physiological system drives it.
5. Make abstention a first-class behaviour under motion or poor signal quality.
6. Enforce honesty structurally, so no clinical or diagnostic claim can be emitted.
7. Deliver an interface a non-expert can understand, including the system’s own limits.

4 Project Planning and Timeline

I decomposed the problem into six dependent concerns—data and contracts, feature engineering, modelling, runtime, serving, and validation/honesty—and adopted a deliberate *software-first* strategy: prove the entire pipeline on public datasets and a synthetic sensor stream before attaching hardware, so that model, backend, and interface are all tested against the exact protocol the device will later use. A fixed raw-event schema defined up front guarantees that a synthetic generator, a replayed dataset, and future firmware all speak the same protocol.

Table 1 shows the project timeline as a Gantt chart across ten working weeks; shaded cells denote active weeks.

The two heaviest phases were the runtime (Phase 5, where personalization and abstention live) and the accuracy work (Phase 8, which raised the model from 0.831 to 0.919). Phase 7 produced an important negative result—that naive dataset pooling harms accuracy—which I acted on by abandoning pooling.

5 Methodology

5.1 Windowing

I slide a 60-second analysis window over the aligned signals, stepping every 30 seconds. Each window yields one fixed-schema feature row. A window overlapping a sensor dropout receives a low coverage score and is routed to the abstention layer (Section 5.5) rather than silently discarded.

5.2 Feature extraction

From four channels I compute 19 absolute features per window using NeuroKit2 [4] for pulse-peak detection and electrodermal decomposition. These include time-domain heart-rate-variability statistics, frequency-domain HRV, and a convex-optimization decomposition of skin conductance into tonic and phasic components [3]. The full feature set is enumerated in Section 7. Two features quantify sympathovagal balance and the low/high-frequency ratio of the inter-beat interval spectrum, defined as

$$\text{LF/HF} = \frac{\int_{0.04}^{0.15} S(f) df}{\int_{0.15}^{0.40} S(f) df}, \quad (1)$$

where $S(f)$ is the power spectral density of the inter-beat-interval series; a higher ratio indicates greater sympathetic dominance.

5.3 Personalization

This is the core of the method. Let x_i be a physiological feature for window i of a given user, and let $\mathcal{B}$ be the set of windows in that user’s initial quiet calibration period (the first ~ 300 seconds of accepted data). I compute the personal mean and standard deviation

$$\mu = \frac{1}{|\mathcal{B}|} \sum_{j \in \mathcal{B}} x_j, \quad \sigma = \sqrt{\frac{1}{|\mathcal{B}|} \sum_{j \in \mathcal{B}} (x_j - \mu)^2}, \quad (2)$$

and append the personalized feature

$$x_i^{(p)} = \frac{x_i - \mu}{\max(\sigma, \varepsilon)}, \quad (3)$$

where ε is a small floor guarding against division by zero. Each physiological feature is thus represented *twice*—in absolute units and as a personal deviation—doubling the effective feature count to 36. Crucially, the reference window $\mathcal{B}$ is defined in *seconds of signal*, not a fixed count of windows, so that the calibration performed live by the deployed system exactly matches the reference used during training, regardless of window step.

5.4 Model architecture

I use gradient-boosted decision trees (a histogram-based gradient boosting classifier [5]), a deliberate choice over deep learning. With only 15 training subjects and tabular features, tree ensembles are the appropriate and best-performing tool at this data scale; they handle missing values natively (a dropped sensor simply becomes a NaN, requiring no imputation); and they remain interpretable. The model has two heads that share the same 36 inputs:

- **Head A—graded stress.** A binary classifier (baseline vs. stress) wrapped in isotonic probability calibration [6], so its output is a genuine probability p . The graded index is $\text{index} = \lfloor 100p \rfloor$, with ordinal bands *calm* ($p < 0.45$), *elevated* ($0.45 \leq p < 0.70$), and *high* ($p \geq 0.70$).
- **Head B—affect.** A four-class classifier (baseline / stress / amusement / meditation) that separates stress arousal from positive arousal, since a raised heart rate during an enjoyable moment is not stress and a single binary model cannot distinguish the two.

Calibration matters: without it a classifier’s raw output is a score, not a probability, and the 0–100 index would be arbitrary. Isotonic calibration fits a monotonic mapping from scores to empirical probabilities, making the index interpretable.

5.5 Abstention

Before the model is consulted, a quality gate inspects motion and signal coverage. If dynamic wrist motion exceeds a threshold, or coverage / pulse-signal quality fall below thresholds, the model is *not invoked* and the window is reported as `motion_paused` or `poor_signal`. On abstention the graded index, level, and affect state are all null. This is a designed feature: a photoplethysmographic reading during hand movement reflects motion artefact rather than blood volume, and refusing to score is strictly preferable to emitting a confident but meaningless number.

5.6 Evaluation protocol

All reported metrics use grouped leave-one-subject-out cross-validation: for each fold one subject is entirely withheld, the model is trained on the remainder, and it is scored only on the withheld subject. I report balanced accuracy—the mean of per-class recalls—which is robust to class imbalance:

$$\text{BA} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right), \quad (4)$$

compared against a majority-class dummy baseline of 0.5.

5.7 Honesty enforcement

A regression test fails the build if any user-facing string contains clinical terminology (e.g. “panic,” “anxiety,” “burnout,” “diagnosis,” “clinical,” “medical”), making over-claiming structurally impossible rather than a matter of discipline.

6 Datasets

I used three datasets with strictly separated roles, summarized in Table 2.

I trained *only* on WESAD. My own experiments confirmed the literature: naively pooling WESAD with Stress-Predict lowered LOSO accuracy from 0.83 to 0.62, so I abandoned pooling and retained the other two datasets strictly as external validation. WESAD’s labels derive from a validated protocol, providing trustworthy ground truth; no equivalent labelled dataset exists for conditions such as burnout, which is one reason the system does not attempt to predict them.

Table 2: Datasets and their roles. Only WESAD was used for training.

Dataset	Subjects	Role	Notes
WESAD [1]	15	Training	Validated laboratory stress protocol. I used only the wrist-worn channels and discarded the cleaner chest sensors deliberately, since the target device has no chest strap. ~ 919 labelled windows after quality filtering.
Stress-Predict [7]	35	Held out	Different people, a different lab stress protocol; used to probe generalization, never trained on.
Nurse Stress [8]	15	Held out	Real hospital shifts with sparse self-report labels—the messiest, most naturalistic data; never trained on.

Table 3: The 19 absolute features per window, by sensing channel. Each physiological feature (all except `ppg_quality` and `valid_fraction`) additionally contributes a personalized $(\cdot)^{(p)}$ counterpart, for 36 inputs total.

Channel	#	Features
Pulse / PPG	7	mean heart rate; two HRV measures (SDNN-like, RMSSD); pulse-signal quality; LF power; HF power; LF/HF ratio
Skin conductance / EDA	7	tonic level; slope; count and mean amplitude of skin-conductance responses; tonic mean; tonic slope; phasic mean (cvxEDA)
Skin temperature	2	mean; trend (negative sign in the arousal direction—stress <i>cools</i> the skin)
Motion / accelerometry	2	dynamic motion energy (RMS); 95th percentile (also the abstention trigger)
Coverage	1	fraction of the window with usable samples

7 Inputs

Each 60-second window yields **36 model inputs**: 19 absolute features plus 17 personalized counterparts. The two quality/coverage features are not personalized, as they describe the recording rather than the person. Table 3 enumerates the absolute features by channel.

Two properties of this input design are worth emphasizing. First, EDA contributes 7 of the 19 features—tied with pulse for the largest block—and is the purest arousal signal, yet is the channel most often absent on low-cost hardware. Second, the thermal features carry a *negative* sign in the arousal direction: because stress cools the skin, a falling temperature *increases* the modelled arousal, a subtlety that a naive “higher is worse” design would get backwards.

8 Results

8.1 Accuracy

Table 4 reports the final model under grouped LOSO on WESAD.

Head A additionally attains a macro-F1 of 0.918. For context, the strongest comparable wrist-only LOSO result in the literature is 0.874 [2]; the system exceeds it.

Table 4: Final model performance under grouped leave-one-subject-out on WESAD.

Head	Task	Balanced acc.	Chance
A	Graded stress (binary)	0.919	0.500
B	Affect (4-class)	0.616	0.250

Table 5: Development of Head A. Every row is grouped LOSO on WESAD.

Configuration	Balanced acc.
Baseline: 13 features, logistic regression	0.831
Naive pooling of WESAD + Stress-Predict (<i>rejected</i>)	0.617
Drop pooling—train on WESAD only	0.831
+ frequency-domain HRV and cvxEDA features	0.880
+ per-subject baseline personalization	0.919

8.2 Ablation: how the accuracy was reached

Table 5 traces the model’s development. Every row is measured identically (WESAD LOSO), so the steps are directly comparable.

The two gains that survived scrutiny were the richer feature set (+0.049) and personalization (+0.039). The rejected pooling configuration is retained in the table because it is instructive: adding data *lowered* accuracy, exactly as the literature predicts for cross-protocol pooling without domain adaptation.

8.3 Honest limits

I report three limitations directly inside the deployed interface, not merely here.

- **The error bar is non-trivial.** With only 15 subjects, the standard error on the balanced accuracy is ± 0.036 .
- **The model is good on average, not uniformly.** Per-subject balanced accuracy ranges from 0.50 (chance) to 1.00. I therefore present the index as a personal *trend* over time, never as a precise instantaneous measurement.
- **0.919 is the ceiling for this data.** After reaching it, I ran a systematic tuning sweep over window size, prediction smoothing, seed ensembling, calibration length, and boosting hyperparameters. Nothing improved on 0.919 with statistical significance. In the process I caught two errors in my own work—a bug in which a calibration parameter silently changed with window size, and a case of selecting the best configuration on the same test folds used to report it—and I documented both as a formal negative result. The binding constraint is the dataset size, not the model.

8.4 System results

The full automated test suite (over 300 tests) passes, as does an end-to-end software acceptance gate. The backend genuinely streams: I verified 18 windows arriving one at a time over a live WebSocket connection, correctly tracing the arc from calm through high stress, then `motion_paused`, then `poor_signal`. The interface was verified in a real browser: a first-time user is presented with a single clear action, watches the reading accumulate live, and sees the model’s abstention states and error bar rendered honestly rather than hidden.

9 Discussion and Limitations

The results support a clear thesis: at this data scale, *how* the signal is represented matters more than model sophistication. Personalization and a physiologically motivated feature set together account for the entire improvement from 0.831 to 0.919, while an extensive hyperparameter search yielded nothing. This is a useful corrective to the reflex of reaching for larger models; with 15 subjects, the path to higher accuracy runs through more subjects and better labels, not deeper networks.

The most consequential limitation is dataset size. Fifteen training subjects impose a ± 0.036 standard error and a wide per-subject spread, and no amount of modelling can escape it. A second limitation is that Head B’s affect classes derive from a single laboratory protocol, so its distinction between stress and positive arousal should be read as suggestive rather than definitive. A third is scope: the system measures *physiological arousal relative to baseline*, which is what the sensors can support, and deliberately makes no claim about specific affective disorders, because the peripheral signature of anger, fear, and excitement is largely shared and cannot be disambiguated from these channels alone.

10 Hardware and Deployment

The intended device integrates a MAX30102 optical pulse sensor, a BNO inertial measurement unit (motion), and an MLX90614 infrared skin-temperature sensor, covering three of the four channels. It currently *lacks a skin-conductance (EDA) sensor*, which accounts for 7 of the 19 features and—as established above—is the cleanest arousal signal. Adding that sensor is the next hardware step, and I am quantifying its value by retraining the model without EDA on the same data, so the decision is driven by a measured accuracy cost rather than intuition.

Because the pipeline is built around a fixed raw-event contract, hardware only ever replaces the *data source*: features, model, explanations, and interface are identical whether data arrive from a recording, a synthetic generator, or a live board. For demonstration, a held-out recording is streamed through the exact path the hardware will use, so that the pipeline validated in software is the pipeline the device runs.

11 Conclusion and Future Work

I set out to convert low-cost physiological signals into an honest, personalized, real-time stress estimate. The system attains a balanced accuracy of 0.919 under the strictest available evaluation—above the published wrist-only benchmark—built on a personalization method that adapts to each individual. Equally important, it is candid about its limits: it abstains when it cannot see, it reports its own uncertainty, and it is structurally barred from making clinical claims. The clearest avenues for future work are (i) adding the missing skin-conductance sensor and measuring its contribution, (ii) re-scoring the two held-out datasets against the current model to quantify real-world generalization, and (iii) enlarging the training population, which the evidence indicates is the only route past the current accuracy ceiling.

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